Warm Up Activity: Unit 6, Momentum I

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1 Memory Bank

- $P = W/\Delta t$ ... Definition of power, ratio of work done to time duration
- $\vec{p} = m\vec{v}$ ... Definition of momentum
- $\vec{p}_{f,\text{tot}} = \vec{p}_{i,\text{tot}}$ ... Momentum conservation

2 Power

1. Suppose you are on a camping trip. (a) If your body burns 120 W as you sit comfortably and enjoy nature, how many kcal do you need to replace the energy spent after 2 hours? (b) Now suppose you were *shivering* the whole time, because you need warmer clothing. Shivering requires 425 W. How many kcal do you need to replace the energy spent?

3 Momentum and Impulse

1. A gas molecule has a mass of 20×10^{-25} kg and an average speed of 350 m/s (Fig. 1). What is the magnitude of the momentum in kg m/s?

2. Suppose this molecule collides with the side of the glass beaker, turns around, and flies off in exactly the opposite direction at the same speed. What is the *change* in momentum, $\Delta\vec{p} = \vec{p}_f - \vec{p}_i$? (This is how we build up the **kinetic theory of gases**).

3. A common injury in car accidents is a neck inury, because the head is snapped back and forward during the impact. (a) Show algebraically that $\vec{F}_{\text{net}} = \Delta\vec{p}/\Delta t$. (b) Given that $\vec{F}_{\text{net}} = \Delta\vec{p}/\Delta t$, we see that $\vec{F}_{\text{net}}\Delta t = m\Delta\vec{v}$. The quantity $\vec{F}_{\text{net}}\Delta t = \Delta\vec{p}$ is called *impulse*. (c) Suppose the head weighs 4 kg, and the initial velocity of the head is 12.5 m s^{-1}, but the car crash brings it to a stop in 50 ms. What is the net force and the acceleration of the head?

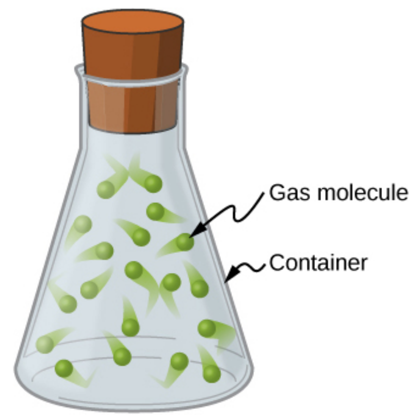

Figure 1: A beaker full of gas molecules.

4 Momentum Conservation

1. Two molecules collide and stick together, forming one larger molecule. Each molecule weighs 20×10^{-25} kg. One has a velocity of 350 m/s, and the other has a velocity of -350 m/s. (a) What is the total initial momentum (adding the two momenta)? (b) What is the final speed of the big new molecule?